2024-25 First Semester
MATH1053 Linear Algebra I

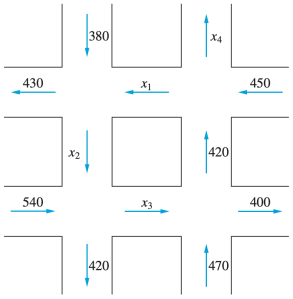
Assignment 3

Due Date: Due on 8/Oct/2024 (Tuesday), at the end of lecture.

- Write down your **CHN name** and **student number**. Write neatly on **A4-sized** paper (*staple if necessary*) and **show your steps**.
- **Late submissions or answers without steps won't be graded.**

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1. In the downtown section of a certain city, two sets of one-way streets intersect as shown in the following figure. The average hourly volume of traffic entering and leaving this section during rush hour is given in the diagram. Determine the values of x_1, x_2, x_3 and x_4 for the following traffic flow diagram, using Gauss-Jordan elimination. *Hint: Set up a system of linear equations. One equation for each intersection.*



2. Consider a linear system whose augmented matrix is of the form

$$\left(\begin{array}{ccc|c} 1 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \\ 1 & 3 & a & b \end{array}\right)$$

   (a) For what values of a and b will the system have infinitely many solutions?

   (b) For what values of a and b will the system be inconsistent?

3. Write out every possible reduced row echelon form for a 3×3 matrix. Make sure you explain carefully why your answer includes *all* possibilities.

   For example, a rref of a 3×3 matrix could be the one with 2 pivots: $\begin{pmatrix} 1 & 0 & * \\ 0 & 1 & * \\ 0 & 0 & 0 \end{pmatrix}$.

4. Let
$$A = \begin{pmatrix} 2 & 3 & 6 & 4 \\ 4 & 7 & 2 & 5 \\ 0 & 3 & 8 & 7 \end{pmatrix}, B = \begin{pmatrix} 2 & 3 & 6 & 4 \\ 0 & 1 & -10 & -3 \\ 0 & 3 & 8 & 7 \end{pmatrix}, C = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \\ 7 & 8 \end{pmatrix} \text{ and } D = \begin{pmatrix} 1 & 20 \\ 3 & 40 \\ 5 & 60 \\ 7 & 80 \end{pmatrix}.$$
a) Find an elementary matrix E such that $EA = B$.
b) Find an elementary matrix F such that $CF = D$.
c) Write down the inverses of E and F.

5. Find the inverse of matrix A and B using elementary row operations on $[A|I]$ and $[B|I]$, respectively.
(a) $A = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}$; (b) $B = \begin{pmatrix} 0 & 1 & 3 \\ 0 & 2 & 1 \\ -2 & 3 & 5 \end{pmatrix}$.

6. Prove that if A is row equivalent to B and B is row equivalent to C, then A is row equivalent to C.

7. Suppose A, B are both $n \times n$ matrices. Prove AB is nonsingular if and only if both A and B are nonsingular.

8. Let A be a 3×3 matrix and suppose that $2\mathbf{a}_1 + \mathbf{a}_2 = 4\mathbf{a}_3$, where $\mathbf{a}_j$ is the jth column of A. How many solutions does the system $A\mathbf{x} = \mathbf{0}$ have? Explain. Is A nonsingular? Explain. (*Hint: Rewrite* $2\mathbf{a}_1 + \mathbf{a}_2 - 4\mathbf{a}_3 = \mathbf{0}$ *as a matrix equation and analyze it.*)

9. Let
$$A = \begin{pmatrix} O & A_1 \\ A_2 & O \end{pmatrix}$$
where all four submatrices are $k \times k$. Determine A^2 and A^4.

10. True or false. If false, give a counterexample. If true, explain or prove your answer briefly.
(a) If matrix A is in rref, then at least one of the entries in each column must be 1.
(b) The product of two $n \times n$ elementary matrices is an elementary matrix.
(c) If $AB = O$, then $BA = O$
(d) If $AC = BC$ and $C \neq O$ (the zero matrix), then $A = B$.